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XII - CHEMISTRY

MODEL QUESTION PAPER - 2

Max. Marks: 70 Duration: 3 Hrs

Part - I

Answer all the questions. Choose the correct answer. [15 x 1 = 15]

1. Aluminium is extracted from alumina (Al_2O_3) by electrolysis of a molten mixture of ——.
(a) $Al_2O_3 + KF + Na_3AlF_6$ (b) $Al_2O_3 + HF + NaAlF_4$
(c) $Al_2O_3 + CaF_2 + NaAlF_4$ (d) $Al_2O_3 + Na_3AlF_6 + CaF_2$

2. Oxidation state of carbon in its hydrides ——.
(a) +4 (b) -4 (c) +3 (d) +2

3. Assertion (A): Xenon is used in high speed electronic flash bulbs used by photographers.
Reason (R): Xenon emits an intense light in discharge tubes instantly.
(a) A and R are correct and R explains A.
(b) A and R are correct but doesn't explains A
(c) A is correct but R is wrong (d) A is wrong but R is correct.

4. Which one of the following elements show high positive electrode potential?
(a) Ti^{2+} (b) Mn^{2+} (c) Co^{2+} (d) Cr^{2+}

5. Solid NH_3, solid CO_2 are examples of ——.
(a) covalent solids (b) polar molecular solids.

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(c) molecular solids (d) ionic solids.

7. After 2 hours, a radioactive substance becomes $\left(\frac{1}{16}\right)^{th}$ of original amount. Then half life is (in mins) —.

(a) 60 mins (b) 120 mins (c) 30 mins (d) 15 mins

8. As the temperature increases, the pH of KOH solution —.

(a) will decrease (b) will increase (c) remains constant

(d) depends upon concentration of KOH solution.

9. Which one of the following on hydrolysis give galactose and glucose?

(a) Maltose (b) Cellulose (c) Lactose (d) Sucrose

10. In the following sequence of reactions,

$$CH_3-CH_2OH \xrightarrow{P+I_2} A \xrightarrow{Mg/ether} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$$

The compound D is —.

(a) Butanal (b) n-butyl alcohol (c) propan-1-ol

(d) Propanal.

11. The method by which aniline cannot be prepared is —.

(a) degradation of benzamide with Br_2/NaOH.

(b) potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution.

(c) reduction of nitrobenzene with $LiAlH_4$.

(d) reduction of nitrobenzene by Sn/HCl.

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12. Assertion (A) : Copper sulphate can be stored in a zinc vessel.

Reason (R) : Zinc is less reactive than copper

(a) Both A and R are correct (b) Both A and R are wrong (c) A is correct but R is wrong.

(d) A is wrong but R is correct.

13. Fog is a colloidal solution of ——.

(a) solid in gas (b) gas in gas (c) liquid in gas

(d) gas in liquid.

14. Formula of tris (ethane-1,2-diamine) iron (II) phosphate

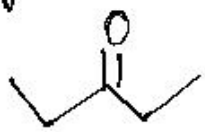
(a) $[Fe(CH_3-CH(NH_2)_2)_3](PO_4)_3$

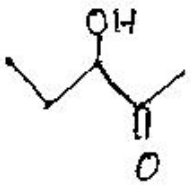
(b) $[Fe(H_2N-CH_2-CH_2-NH_2)_3](PO_4)$

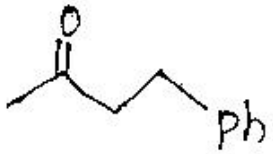
(c) $[Fe(H_2N-CH_2-CH_2-NH_2)_3](PO_4)_2$

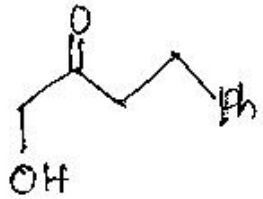
(d) $[Fe(H_2N-CH_2-CH_2-NH_2)_3](PO_4)_2$.

15. In which case chiral carbon is not generated by reaction with HCN

(a) 

(b) 

(c) 

(d) 

Part - II

Answer any six questions. Question No: 22 is compulsory.

16. List out the common refining methods.

17. Mention the use of boric acid.

18. Which metal in 3d series exhibits +1 oxidation state

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most frequently and why?

19. How many lattice points are there in one unit cell of each of the following lattice?
(i) Face-centred cubic (ii) Body centered.

20. The aqueous solution of sugar does not conduct electricity whereas when sodium chloride is added to water, it conducts electricity. Justify this statement.

21. What are the limitations of Freundlich adsorption isotherm?

22. Write the chemical equation for Williamson synthesis of 2-ethoxy-2-methyl pentane starting from ethanol and 2-methyl pentan-2-ol.

23. Aniline is basic in nature. Justify this statement.

24. What happens when a colloidal sol of $Fe(OH)_3$ and As_2S_3 are mixed?

Part - III

Answer any six questions. Question No: is compulsory.

25. Explain about fullerenes.

26. Explain about the effects of lanthanide contraction.

27. The ionization constant of acetic acid is 1.74×10^{-5}. Calculate the degree of dissociation of acetic acid in its 0.05 M solution. Calculate the concentration of acetate ions in the solution and its pH.

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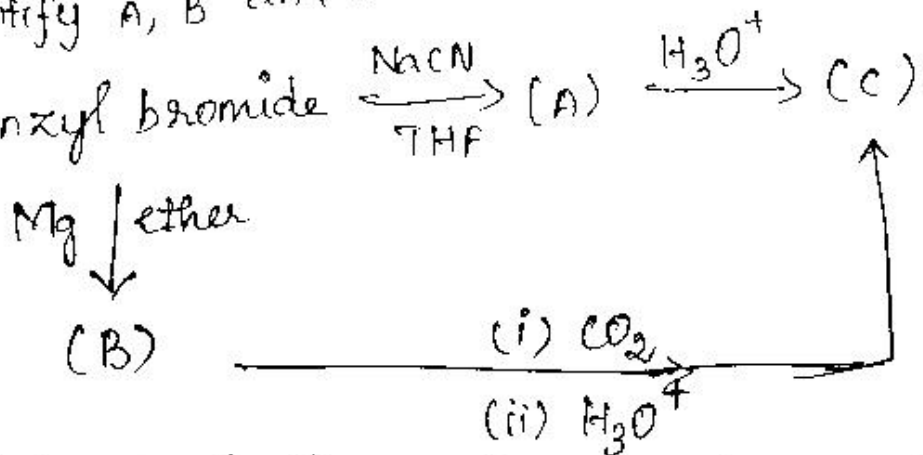
28. Describe the action of active centres present in the catalyst.

29. The electrochemical cell reaction of the Daniel cell is $Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$

What is the change in the cell voltage on increasing the ion concentration in the anode compartment by a factor 10?

30. What are the factors which influence the adsorption of a gas on a solid?

31. Identify A, B and C.



32. Explain about the cyclic structure of glucose.

33. How will you distinguish between primary, secondary and tertiary aliphatic amines.

Part - IV

34. (a) (i) Bleaching action of chlorine is permanent. Justify this statement.

(ii) What is Royal water?

[OR]

(b) (i) What is lanthanide contraction and what are the effects of lanthanide contraction?

(ii) How many series are in d-block elements? What are they?

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35. (a) (i) Write the postulates of Werner's theory.

(ii) What is coordination entity? Give example.

[OR]

(b) (i) Distinguish between hexagonal close packing and cubic close packing.

(ii) What are molecular solids? Explain their classification with suitable examples.

36. (a) Derive an expression for Ostwald's dilution law

[OR]

(b) (i) For strong electrolytes the molar conductivity increases on dilution and reaches a maximum value at infinite dilution. Justify this statement.

(ii) A solution of silver nitrate is electrolysed for 20 minutes with a current of 2 amperes. Calculate the mass of silver deposited at the cathode. (3)

37. (a) (i) Write a note about negative catalyst. (2)

(ii) What is gold number?

(OR)

(b) An organic compound (A) of molecular formula C_2H_6O on reaction with Con. H_2SO_4 at 443K gives an unsaturated hydrocarbon (B). (B) on reaction with Bayer's reagent produces (C) of molecular formula $C_2H_6O_2$. (C) on reaction with anhydrous $ZnCl_2$ produces (D) of molecular formula C_2H_4O. (D) reduces Tollen's reagent. Identify A, B, C and D. Explain the reactions.

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38. (a) (i) Arrange the following in increasing order of basic strength

(a) aniline, p-toluidine and p-nitroaniline.

(b) $C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5NH_2$, p-$Cl-C_6H_4-NH_2$

(ii) Convert ethanamine into methanamine.

[OR]

(b) (i) Identify X and Y.

$CH_3COCH_2CH_2COOC_2H_5 \xrightarrow{CH_3MgBr} X \xrightarrow{H_3O^+} Y$

(ii) Explain the action of the following reactants with benzoic acid.

(i) $SOCl_2$ (ii) Br_2 / $FeBr_3$

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